Journal of Science and Medicine in Sport 20 (2017) 549–554

Contents lists available at ScienceDirect

Journal of Science and Medicine in Sport

journal homepage: www.elsevier.com/locate/jsams

Original research

The "Strengthen your ankle" program to prevent recurrent injuries: A randomized controlled trial aimed at long-term effectiveness☆

M. Van Reijen[a,b], I. Vriend[a,b,c], V. Zuidema[c], W. van Mechelen[a,b,d,e,f], E.A. Verhagen[a,b,f,g,*]

[a] Amsterdam Collaboration on Health and Safety in Sports, IOC Research Center, VUmc/AMC, The Netherlands
[b] Department of Public & Occupational Health, EMGO Institute for Health and Care Research, VU University Medical Centre, The Netherlands
[c] Consumer Safety Institute VeiligheidNL, The Netherlands
[d] School of Public Health, Physiotherapy and Population Sciences, University College Dublin, Ireland
[e] School of Human Movement and Nutrition Sciences, Faculty of Health and Behavioural Sciences, University of Queensland, Australia
[f] UCT/MRC Research Unit for Exercise Science and Sports Medicine (ESSM), Department of Human Biology, Faculty of Health Sciences, University of Cape Town, South Africa
[g] Australian Centre for Research into Injury in Sport and its Prevention, Federation University Australia, Australia

ARTICLE INFO

Article history:
Received 28 April 2016
Received in revised form
19 November 2016
Accepted 3 December 2016
Available online 9 December 2016

Keywords:
Injury prevention
Ankle injury
E-health

ABSTRACT

Objectives: Recurrent ankle sprains can be reduced by a neuromuscular training program (NMT). The way NMT is delivered may influence the incidence of long term recurrent injuries, residual pain and disability.
Design: This RCT with a follow-up of twelve months, evaluated whether the implementation method of a proven effective NMT program delivered by a mobile application or a written instruction booklet, resulted in differences in injury incidence rates, functional ankle disability/pain in the long term, assuming equal compliance – as is shown in previous research – with the 8-week intervention.
Methods: 220 athletes with a history of ankle sprain were recruited for this RCT. 110 athletes were offered the freely available "Strengthen your ankle App" and the other 110 received a printed Booklet. Primary outcome measure was incidence density of ankle sprains. Secondary outcome measures were residual pain/disability and the individual cumulative number of ankle sprains during follow-up.
Results: The incidence densities of self-reported ankle sprain recurrences were not significantly different between both groups (HR 1.06; 95% CI 0.76–1.49). Median FADI (Functional Ankle and Disability Index) scores increased equally over time in both groups, indicating a lower rate of limitation and pain in both groups at follow-up. Neither FADI scores nor cumulative recurrent injuries were significantly different between groups.
Conclusions: This study showed that the implementation method of a NMT program by using an App or a Booklet did neither lead to different injury incidence rates in the long term nor did it influence residual functional disability/pain. Assuming equal compliance during the 8-week intervention, both methods show similar effectiveness in twelve-month follow-up.
© 2016 Sports Medicine Australia. Published by Elsevier Ltd. All rights reserved.

1. Introduction

Despite on-going societal and scientific attention, sports injuries continue to pose a substantial health care problem. Although exercise is generally seen as contributing to overall health, the manifestation of a sport or physical activity related injury causes harm both from a personal as well as a societal viewpoint.[1] The ankle is the second most injured body part across all sports and activities, with an ankle sprain being the most common ankle injury.[2] In the Netherlands alone, in 2013, on a population of 17 million 480,000 people suffered from an ankle sprain due to sports.[3]

Previous studies have shown the effectiveness of neuromuscular training programs.[4] Such programs require individuals to complete a multi-week exercise program that should be carried out multiple times a week. Although the effectiveness of such preventive programs has been shown in controlled studies, the majority of individuals involved were neither able nor willing to comply with the prescribed program. Performing only part of such preventive

☆ Trial registration: The Netherlands National Trial Register NTR 4027. The NTR is part of the WHO Primary Registries.

* Corresponding author.
E-mail address: e.verhagen@vumc.nl (E.A. Verhagen).

http://dx.doi.org/10.1016/j.jsams.2016.12.001
1440-2440/© 2016 Sports Medicine Australia. Published by Elsevier Ltd. All rights reserved.

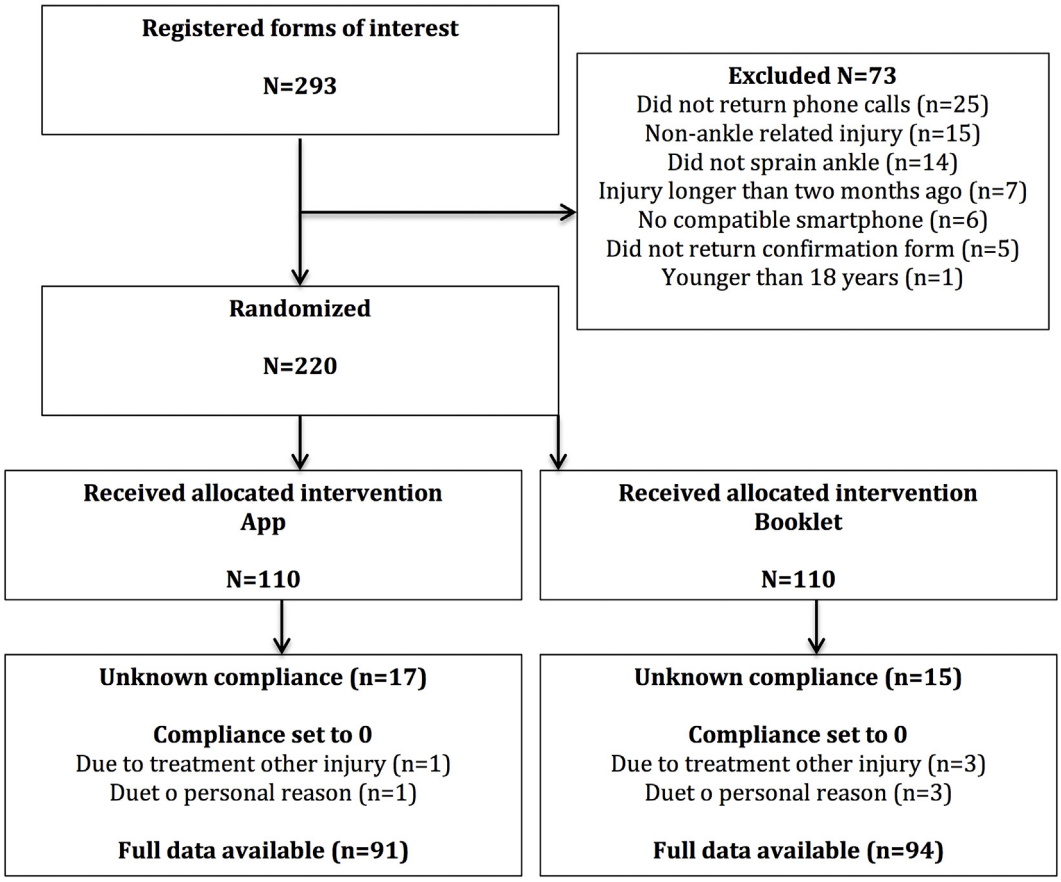

Fig. 1. Flowchart of participant recruitment and follow-up.

exercise programs has been shown to be ineffective in reducing the risk of an (recurrent) ankle sprain.[5,6] In recognition of the major problem that ankle sprains continue to pose for those involved in physical activities and sports, it is, therefore, of importance that available programs are enhanced in such a way that compliance with the exercise scheme is increased.

In order to increase compliance, it is crucial that the method of implementation is seen as attractive and has low barriers of use to the ones involved. The choice of implementation method may influence both compliance and, more importantly, the recurrence of sport injuries. A previous study has shown that during the actual course of an intervention program, there was no difference in compliance when a printed instruction Booklet was used when compared with a mobile application, and that during the course of the intervention program the number of recurrent injuries did not differ between both implementation methods.[7] However, difference in preventive effectiveness may be seen over a longer follow-up period, i.e. during a follow-up period beyond the actual intervention program. It is possible that, although rate of compliance was similar between both methods, the quality of compliance – i.e. the way the exercises were performed – between the App and the Booklet differs. As it is impossible to measure the quality of compliance, we used the severity and total number of recurrent injuries as a measure of compliance quality. The self-reported injuries include all minor sprains and giving-ways. Many of those injuries have little to no consequence. However, injuries that lead to time loss and/or costs can be seen as injuries with a greater severity. Consequently, the aim of the current study was to investigate whether delivering a proven effective neuromuscular training program through a Booklet or an App results in differences in recurrent ankle sprain incidence over a 12-month follow-up, under the condition of equal compliance/adherence *rates* across deliverance methods during the actual 8-week intervention program. A secondary aim was to investigate differences in pain and functional outcomes during daily activities over the 12 months of follow-up. The answer to these questions is of importance to further develop the methods of delivery of the neuromuscular training program in particular and injury prevention in general, which are both effective and attractive for those involved.

2. Methods

This study was a randomized controlled trial. A detailed description of the study protocol has been published previously.[8] The Medical Ethics Committee (2013/248) of the VU University Medical Center Amsterdam (VUmc) the Netherlands approved the study design and informed consent procedure for this study. The trial is registered in the Netherlands Trial Registry (NTR4027).

From October 2013 to April 2014 physiotherapy and sport physician practices, national sport federations' websites, digital newsletters and social media were used to recruit active sports participants (athletes) between 18 and 70 years. To be included in the study athletes had to have incurred a self-reported ankle sprain within two months before inclusion into the study, and had to have access to a mobile smart phone (either Android or iOS). When respondents had suffered a different injury in the same ankle in the preceding year (e.g., fracture of the ankle) they were considered non-eligible. The first author assessed all reported ankle sprains from the participants orally to confirm inclusion eligibility. The flow of participants is shown in Fig. 1.

Previous to the current study on the long-term effects, a study was carried out with the same study sample on the short-term

Table 1
Baseline characteristics of the study population. Numbers are presented as percentage of the population (%), the median with corresponding range or as the mean with corresponding standard deviation (mean ± SD).

	Combined	App	Booklet	*p* Value
Participants (n)	220	110	110	
Number of males (n and %)	110 (50.0)	55 (50.0)	55 (50.0)	1.000
Age (years)	37.9 ± 13.4	37.6 ± 13.1	38.1 ± 13.7	0.741
Weekly sports exposure (hours and range)	3.0 (0–19.5)	3.0 (0–15.0)	3.0 (0–19.5)	0.791
Level of sport (n and %)				0.311
Competitive, international	9 (4,1)	2 (1.8)	7 (6.3)	
Competitive, national	32 (14,5)	19 (17.3)	13 (11.8)	
Competitive, regional	76 (34.5)	40 (36.4)	36 (32.7)	
Recreational, organized	56 (25.5)	25 (22.7)	31 (28.2)	
Recreational, unorganized	47 (42.7)	24 (21.2)	23 (20.9)	
Contact sport (n and %)	136 (61.8)	68 (61.8)	68 (61.8)	1.000
Severity of inclusion sprain (n and %)				0.666
Grade 1	91 (41.3)	47 (42.7)	44 (40.0)	
Grade 2	64 (28.9)	34 (30.9)	30 (27.3)	
Grade 3	18 (8.0)	10 (9.1)	8 (6.9)	
Unknown	48 (21.8)	19 (17.3)	29 (26.4)	
Medically treated (n and %)	134 (61.0)	67 (61.0)	65 (59.0)	0.783

effects. The primary outcome measure of this short-term study was compliance to the 8-week training program. Therefore, sample size calculations were based upon the expected compliance rates and grounded on previously reported compliance rates to the same program.[11] Full compliance rates in the control group (Booklet) were expected to be around 25%. A doubling of this rate to at least 50% in the intervention group (i.e., the App group; see below) was considered to be clinically relevant. Based upon a beta of 0.90 and an alpha of 0.05, a total of 158 athletes was required, divided across both study groups. Assuming a dropout rate of 20%, a sample of 190 participants was calculated.

After participants had finished usual care for their ankle sprain treatment, and had returned the informed consent and baseline questionnaire, they were randomly assigned to one of two study groups; i.e. an App group and a written (print) materials only group (i.e. 'Booklet group'). Participants were allocated to the study groups through a random number generator. The App group was instructed to use the "Strengthen your ankle" (in Dutch: "Versterk je Enkel") App on their mobile phone, which could be downloaded for free from either the App Store[9] or the Google Play store.[10] The Booklet group received a Booklet with the same NMT program on paper.

All participants received the same balance board (machU/MSG Europe BVBA). Both the Booklet and the App contained the same NMT program that consisted of 6 different exercises to be performed during 3 sessions a week, for a period of 8 weeks. Difficulty and load of the exercises were prescribed to increase every week. The NMT program has been shown to be cost-effective in reducing recurrent ankle sprain in previous trials.[5,12] The App provided the participant with an interactive exercise schedule, possibility for reminders, written instructions, and narrated video instructions on each exercise. The Booklet used pictures and written instructions to explain the exercises. During the 8 weeks of the intervention, weekly questionnaires were sent to all participants. After the 8 weeks these questionnaires were sent monthly for a further 10 months. During these 10-month follow-up participants were no longer required to follow the NMT program. These monthly questionnaires included questions on recurrent injuries, exposure to sport activities and, pain and limitations as a result of the initial and/or recurrent ankle sprain. The study showed no differences in compliance with the intervention between the App-group and the Booklet-group during the 8 weeks of the intervention program.[7]

An online baseline questionnaire was used to collect participants' information on physical characteristics, injury and sports history, use of preventive measures (tape, brace), and severity, received treatment and rehabilitation of the inclusion ankle sprain.

The primary outcome measure of the current study was incidence density of ankle sprains as measured by self-report during the twelve-month follow-up. Injury incidence density (IID) was defined as the number of recurrent ankle sprains per 1000 h of sports exposure. During the twelve-month follow-up, participants were asked monthly through online questionnaires about their hours of sports exposure and whether they had sustained an ankle sprain in the previous month. Ankle sprain recurrence was defined as a self-reported inversion moment of the same ankle, by which both a mere moment of inversion (giving way) as well as clinical ankle sprains were included. Self-reported recurrent ankle sprains were categorized by severity by looking at recurrences that had led to time-loss or costs. Time loss was defined as the discontinuity of (sport) activity and/or missing (part of) the next planned (sports) activity due to the recurrent ankle sprain.[11] Sprains that resulted in either direct or indirect costs were categorized as sprains leading to costs. These methods have been used previously in 2 comparable studies looking at the effectiveness of the NMT program to prevent recurrent ankle sprain.[6,12]

Secondary outcome measure was the Functional Disability Ankle Index.[13] This index has been validated previously and has been used to describe pain at the ankle and difficulty during daily activities.[14,15] The FADI consists of 26 items scored on a 5 point Likert scale. The FADI was included in the monthly questionnaires. The index score was recorded as a percentage. A full score of 104 points resulted in a 100% score. The higher the FADI, the lower the rate of limitation and pain as a result of the ankle sprain. Twenty-two items questioned limitations on daily activities. Subjects rated the activity as no difficulty at all (4 points), slight difficulty (3 points), moderate difficulty (2 points), extreme difficulty (1 point), unable to do (0 points). Not applicable (N/A) was scored when the activity was limited by something other than the ankle sprain. If N/A was scored, this item was not included in the final score. For pain related items scores were: no pain (4 points), mild pain (3 points), moderate pain (2 points), extreme pain (1 point) and unbearable pain (0 points).

Another secondary outcome measure was the cumulative number of ankle sprains and/or inversions during the twelve months of

Table 2
Injury incidence density (IID) of self-reported, time loss, and ankle sprain recurrences leading to costs (95% CI) per 1000 h of sports participation, as well as the Hazard Ratio (95% CI) derived from Cox regression analyses.

	App		Booklet		HR
	n	IID	n	IID	
Overall					
Self-reported	70	15.59 (11.94–19.24)	69	15.84 (12.10–19.58)	1.06 (0.76–1.49)
Time loss	13	1.50 (0.69–2.32)	25	2.71 (1.65–3.77)	0.55 (0.82–1.09)
Costs	16	1.96 (1.00–2.92)	16	1.85 (0.95–2.76)	1.13 (0.56–2.27)
Male	55		55		110
Self-reported	34	9.88 (6.56–13.20)	27	8.74 (5.00–12.48)	1.12 (0.68–1.87)
Time loss	8	1.49 (0.46–2.52)	11	3.65 (1.49–5.81)	0.51 (0.20–1.29)
Costs	9	1.76 (0.61–2.9)	6	1.99 (0.40–3.58)	1.17 (0.41–3.30)
Female	55		55		110
Self-reported	36	34.25 (23.06–45.44)	42	20.99 (14.57–27.42)	1.09 (0.70–1.72)
Time loss	5	1.52 (0.19–2.85)	14	2.49 (1.19–3.80)	0.51 (0.18–1.44)
Costs	7	2.30 (0.60–4.00)	10	1.61 (0.61–2.61)	1.04 (0.39–2.77)

follow-up from baseline, per participant. This secondary outcome measure was calculated by adding all recurrent injuries that were sustained during the twelve months of follow-up.

Mean baseline differences between the App and the Booklet group were determined using an independent samples *t*-test for continuous data (age, body weight, body height, exposure) and Fisher's exact test (with Monte Carlo 95% confidence interval simulation due to small number of samples) for categorical data (level of sport, severity of ankle sprain, gender). Baseline measurements were based on the total number of participants (n = 220) that were allocated to either one of the interventions.

Multivariable Cox regression analyses compared risk of self-reported, time loss and cost recurrent ankle sprains between the groups. Injury incidence analyses were checked for confounders and effect modifiers: age, gender, type of sport (contact, non-contact), weekly sports exposure (hours), severity of inclusion sprain (grade 1, 2, 3 or unknown), medical treatment (yes/no) and level of sport (competitive international, competitive national, competitive regional, recreational organized and recreational unorganized). During the first 8 weeks of the study, all participants were questioned on their compliance with the exercise program. Performing at least 75% of all exercises was regarded as being compliant with the exercise program.[6] No confounders were found. As gender was found to be an effect modifier for rate of compliance in the short term, separate results were presented for men and women.[7]

A Mann–Whitney *U* test was used to calculate the differences between monthly FADI scores and cumulative injury recurrences between both groups as FADI scores are not normally distributed.

All analyses were conducted according to the intention-to-treat principle and differences were considered significant with a significance level of 0.05.

3. Results

A total of 220 athletes was recruited during the recruitment period October 2013 to April 2014. At baseline, both groups were comparable regarding all variables measured (Table 1).

A total of 20,046 h of sports participation was registered during the 12-month follow-up. In the App group the total exposure was 9397 h, whereas in the Booklet group total exposure was 10,648 h. During 12 months' follow-up, there were 139 recurrent ankle injuries: 70 in the App group and 69 in the Booklet group. Thirty-two injuries led to costs and 38 injuries to time loss. Table 2 shows the injury incidence densities of all recurrent injuries, those leading to time loss and those resulting in costs. Via Cox regression analysis no differences in injury incidence density were found between groups ($p < 0.05$).

Median FADI scores increased over time from 92.9 (62.0–100.0) at baseline to 100 (57.6–100.0) at 12 months' follow-up in the App group and from 93.8 (57.1–100.0) at baseline to 100 (78.9–100.0) at 12 months' follow-up in the Booklet group, respectively. In none of the follow-up months FADI scores were significantly different between groups. From the 3rd month onwards, after the training program had been completed, participants experienced little pain and/or disability from their ankle injury. During the final months of the follow-up, pain and disability scores did, on average, not increase. Table S1 (Supplementary material in the online version at DOI: 10.1016/j.jsams.2016.12.001) shows the FADI scores with their range, standard deviations and mean scores. The numbers refer to the month over which the FADI score was measured, i.e.: in the first months (FADI 1) the mean FADI score for the combined group was 90.97 ± 8.57 with a range of (57.05–100).

A total of 58 participants reported more than one self-reported recurrent ankle injury during the follow up. Twelve participants in the App group and 20 in the Booklet group suffered from 2 ankle sprains in the 12-months follow-up; another 4 (App) and 9 (Booklet) participants recurred 3 ankle sprains, 4 (App) and 2 (Booklet) participants dealt with 4 recurrent ankle sprains, and in both groups 3 participants recurred 5 recurrent ankle sprains. Finally, 1 participant in the App group had 6 self-reported ankle sprains within the twelve months. A total of 11 participants suffered 2 injuries leading to costs; 6 in the App group and 5 in the Booklet group. A further 7 participants dealt with 2 injuries leading to time loss, 3 in the App group and 4 in the Booklet group. The difference in total number of injuries, total number of injuries leading to time loss or total number of injuries leading costs did not reach significance.

4. Discussion

The persistent existence of sport-related injuries has provided researchers and health professionals with a continuing challenge to develop and improve effective interventions.[3] To stimulate injured athletes to comply with such interventions, the programs implemented should be attractive with low barriers to use.[1] In the present-day society, where mobile and tablet usage is rising, E-health is thought to provide new opportunities to increase compliance to effective intervention programs.[17]

Mobile applications allow for new features, such as verbal instructions, video's showing the correct execution of exercises and a calendar function to remind users to do the exercises as advised. As our previous study has shown, an App did not lead to better rates of compliance, compared to the traditional way of delivering

an intervention via printed material in the shape of a Booklet. In this studies compliance was measured as the number of exercises that had been done during the intervention program. However, the previous study did not address effects on injury incidence over a twelve-month period, as this study did. Enabling a longer follow-up period allowed for a more detailed analysis of the quality of compliance with the intervention program. However, this study showed, during the follow-up of twelve months, neither a significant difference in injury incidence density between the two groups nor a difference in injuries leading to costs or time loss. In addition, the FADI scores increased equally in both groups during follow-up and reached a ceiling effect after 3 months. Hereafter, pain and disability scores did not increase during the remainder of the follow-up. The cumulative number of recurrent injuries did not show a significant difference between the App and the Booklet group. It can thus be argued that in addition to a similar rate of compliance, also the quality of executing the exercises is very similar between the App and the Booklet.

These results do not, by any means, show that the use of E-health is of no use in sport injury prevention. It does show that the use of mobile applications is as effective as the use of more traditional printed instructions. Both methods of implementation have received an update after completing our study. This update aims to increase the attractiveness and usability and this may lead to even better compliance/adherence rates, thus potentially leading to even better outcomes. Using clear pictures and/or videos in the updated methods may have allowed for a better execution of the exercises involved.

It would be of great interest to see if the simultaneous and combined use of both methods can further increase both rate of compliance rates and quality of compliance and hence contribute to effective sport injury prevention. When both methods of implementation can be used simultaneously, athletes involved have a diversity of means to use. This allows athletes to choose the method that is most convenient at a certain moment in time and/or that is in line with their personal preferences. They can use videos, pictures or a written description to learn the right execution of the exercises.

It is recognized by others that there is a need to measure quality of compliance in home-based exercises.[18] Commonly, studies use a variety of online, written or verbal questionnaires to discover how much and how well the participants performed the exercises.[19–21] Recently, studies have looked at new technology to find out what participants are doing and how much they are doing. An examples of such a method that has been introduced is live feedback via a digital sensor and an iPad[22] and the use of active game technology such as Nintendo Wii and Xbox Kinect.[23] Both methods show promising results. Participants receive direct feedback on the execution of exercises and are motivated to perform the intervention rightfully. The exercises in the neuromuscular training program involve a whole myriad of different movements. Designing technology that can assess and correct the execution of the exercises will be a future challenge.

The results from this study can be seen as an important contribution to the scientific community. Where for the majority of available Apps the App content is not evidence-based,[16] the "Strengthen your ankle" App uses a proven effective intervention.[5]

In the twelve months of follow-up, athletes were contacted monthly to assess their recurrent injuries and FADI scores. In the questionnaires used, they were also asked whether they had continued using the App and/or the Booklet. It is possible that being part of this study protocol has influenced their use of the intervention. Therefore, in a real life situation, without the stimulus of this research, athletes might be less likely to continue the exercises and as a result develop more recurrent injuries. A further limitation of the current study was the sample size calculation. As this sample size has been calculated according to study compliance – rather than long term effectiveness – it is possible that the current sample size has limited power to detect significant differences in long term outcomes.

5. Conclusion

This study showed that the implementation method of a NMT program by using an App or a Booklet did neither lead to different injury incidence rates in the long term nor did it influence residual functional disability/pain. Assuming equal compliance during the 8-week intervention, both methods show similar effectiveness in twelve-month follow-up.

Practical implications

This study showed that the method of implementing the exercises by using an App or a Booklet did not lead to different injury incidence rates in the long term nor did it influence residual functional disability or pain. Assuming equal compliance during the 8-week intervention – as is shown in previous research – both methods showed similar effectiveness in a twelve-month follow-up. This indicates that both methods can be used interchangeably to reduce the risk of recurrent ankle sprains.

Contributors

EV (e.verhagen@vumc.nl) conceived the research idea, MVR (m.vanreijen@vumc.nl, @miriamvanreijen) and EV have written the protocol, MVR has screened and included patients, performed data analyses and is the main author. IV (i.vriend@vumc.nl), WVM (w.vanmechelen@vumc.nl) and VZ (v.zuidema@veiligheid.nl) contributed to ideas in the protocol. All authors have read and commented on the draft version and approved the final version of the manuscript.

Patient consent

Obtained.

Ethics approval

The study was approved by the medical ethics committee of the VU University Medical Center, Amsterdam, The Netherlands (protocol number 2013/248).

Funding

This study was funded by the Netherlands Organisation for Health Research and Development (ZonMw) grant number 525001003. Balance boards were provided by Disporta and Booklets were provided by VeiligheidNL.

Acknowledgments

We would like to thank the following partners for their collaboration and recruitment of study participants: Royal Dutch Society for Physical Therapy (KNGF), Dutch Society for Physical Therapy in Sports (NVFS), Dutch College of General Practitioners (NHG), Dutch Sports Medicine Society (VSG), Dutch Olympic Committee (NOC*NSF), Zilveren Kruis Achmea (ZKA), and Disporta.

References

1. Verhagen E, Bolling C, Finch CF. Caution this drug may cause serious harm! Why we must report adverse effects of physical activity promotion. *Br J Sports Med* 2015; 49(1):1–2.
2. Fong DT-P, Hong Y, Chan L-K et al. A systematic review on ankle injury and ankle sprain in sports. *Sports Med* 2007; 37:73–94.
3. Consument en Veiligheid. Enkelblessures door sport. Available at: http://www.veiligheid.nl/sportblessures/kennis/cijfers-over-sportblessures. Visited 15 februari 2016.
4. Verhagen EALM, Bay K. Optimizing ankle sprain prevention: a critical review and practical appraisal of the literature. *Br J Sports Med* 2010; 44:1082–1088.
5. Hupperets MDW, Verhagen EALM, van Mechelen W. Effect of unsupervised home based proprioceptive training on recurrences of ankle sprain: randomised controlled trial. *BMJ* 2009; 339:b2684.
6. Verhagen EA, Hupperets MD, Finch CF et al. The impact of adherence on sports injury prevention effect estimates in randomised controlled trials: looking beyond the CONSORT statement. *J Sci Med Sport* 2011; 14:287–292.
7. Van Reijen M, Vriend I, Zuidema V et al. Increasing compliance with neuromuscular training to prevent ankle sprain in sport: does the 'Strengthen your ankle' mobile App make a difference? A randomised controlled trial. *Br J Sports Med* 2016:1200–1205. http://dx.doi.org/10.1136/bjsports-2015-095290.
8. Van Reijen M, Vriend I, Zuidema V et al. The implementation effectiveness of the "Strengthen your ankle" smartphone application for the prevention of ankle sprains: design of a randomized controlled trial. *BMC Musculoskelet Disord* 2014; 15:1–8.
9. "Versterk je enkel App" App store: https://itunes.apple.com/nl/app/versterk-je-enkel/id456001033?mt=8. Last visited 24 November 2015.
10. "Versterk je enkel App" Google play store: https://play.google.com/store/apps/details?id=nl.veiligheid.versterkjeenkel&hl=nl. Last visited 24 November 2015.
11. Hupperets MDW, Verhagen EALM, Heymans MW et al. Potential savings of a program to prevent ankle sprain recurrence: economic evaluation of a randomized controlled trial. *Am J Sports Med* 2010; 38:2194–2200.
12. Janssen KW, van Mechelen W, Verhagen EALM. Bracing superior to neuromuscular training for the prevention of self-reported recurrent ankle sprains: a three-arm randomised controlled trial. *Br J Sports Med* 2014; 48:1235–1239.
13. Martin RL, Irrgang J, Burdett RG. Development of the Foot and Ankle Disability Index (FADI) [abstract]. *J Orthop Sports Phys Ther* 1999; 29:A32–A33.
14. Eechaute C, Vaes P, Van Aerschot L et al. The clinimetric qualities of patient-assessed instruments for measuring chronic instability: a systematic review. *BMC Musculoskelet Disord* 2007; 8(18):6.
15. Hale SA, Hertel J. Reliability and sensitivity of the Foot and Ankle Disability Index in subjects with chronic ankle instability. *J Athl Train* 2005; 40(1):35–40.
16. Van Mechelen DM, van Mechelen W, Verhagen EA. Sports injury prevention in your pocket? Prevention apps assessed against the available scientific evidence: a review. *Br J Sports Med* 2014; 48(11):878–882.
17. Boudreaux ED, Waring ME, Hayes RB et al. Evaluating and selecting mobile health apps: strategies for healthcare providers and healthcare organizations. *Transl Behav Med* 2014; 4(4):363–371.
18. Bollen JC, Dean SG, Siegert RJ et al. A systematic review of measures of self-reported adherence to unsupervised home-based rehabilitation exercise programmes using psychometric properties. *BMJ Open* 2014; 4(6):e005044.
19. Bennell KL, Egerton T, Bills C et al. Addition of telephone coaching to a physiotherapist-delivered physical activity program in people with knee osteoarthritis: a randomised controlled trial protocol. *BMC Musculoskelet Disord* 2012; 13:246.
20. Milne M, Hall C, Forwell L. Self-efficacy, imagery use, and adherence to rehabilitation by injured athletes. *J Sport Rehabil* 2005; 14(2):150–168.
21. Levy AR, Polman RCJ, Clough PJ. Adherence to sport injury rehabilitation programs: an integrated psycho-social approach. *Scand J Med Sci Sports* 2008; 18(6):798–809.
22. Riel H, Matthews M, Vicenzino B et al. Efficacy of live feedback to improve objectively monitored compliance to prescribed home-based exercise therapy-dosage in 15 to 19 year old adolescents with patellofemoral pain – a study protocol of a RC superiority trial (the XRCISE-AS-INSTRUcted-1 trial). *BMC Musculoskelet Disord* 2016; 2(17):242. http://dx.doi.org/10.1186/s12891-016-1103-y.
23. Bower K, Louie J, Landesrocha Y et al. Clinical feasibility of interactive motion-controlled games for stroke rehabilitation. *J NeuroEng Rehabil* 2015; 12:63. http://dx.doi.org/10.1186/s12984-015-0057-x.